An explicit rate for ultraflat real Littlewood polynomials

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Abstract

A real Littlewood polynomial of length N is a polynomial $\sum_{j=0}^{N-1} \epsilon_j z^j$ whose coefficients are all $+1$ or -1. OpenAI's preprint *Ultraflat real Littlewood polynomials* (October 5, 2026) proved that for every $\varepsilon > 0$ and every sufficiently large N there is such a polynomial whose modulus lies between $(1-\varepsilon)\sqrt{N}$ and $(1+\varepsilon)\sqrt{N}$ at every point of the unit circle. Its proof gives no value for "sufficiently large". We give one: for $0 < \varepsilon < 1$, every $N \geq \exp(\exp(\exp(10^{12}\log^2(2/\varepsilon))))$ works. In particular, for every $N \geq \exp(\exp(\exp(10^{12}\log^2 4)))$ the error can be taken to be $2\exp(-10^{-6}\sqrt{\log\log\log N})$. The proof follows the original construction and makes each step explicit. Two steps need new arguments: a hypergraph matching lemma whose constants are explicit in the size of the edges, replacing the Pippenger–Spencer theorem, and an exponential decay estimate showing that the auxiliary polynomials need only logarithmically many variables. The bound is far too large to be practical. Its point is that the construction is effective, with every constant written down.

Status and AI assistance. The proofs in this paper were written with the assistance of AI models and were checked by separate AI proof audits. They have not been reviewed by a human expert or formally verified. Novelty is qualified in Section 1.1.

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1 Introduction

A real Littlewood polynomial of length N has the form $P(z) = \sum_{j=0}^{N-1} \epsilon_j z^j$ with every $\epsilon_j \in \{-1, 1\}$. By Parseval's identity the mean of $|P(z)|^2$ over the unit circle is N, so $\sqrt{N}$ is the natural size of $|P|$. A family of such polynomials is *ultraflat* if $|P(z)|/\sqrt{N}$ tends to 1 uniformly on the circle as the length grows. OpenAI's preprint [9] proved that real Littlewood polynomials can be ultraflat at every sufficiently large length. Its argument fixes many auxiliary objects by qualitative choices, so it gives no bound on how large the length must be for a given accuracy.

This paper makes that argument explicit. Theorem 2.1 says that accuracy ε is reached at every length $N \geq \exp(\exp(\exp(10^{12} \log^2(2/\varepsilon))))$, and Corollary 2.2 restates this as an error of $2\exp(-10^{-6}\sqrt{\log\log\log N})$ at length N. Two steps need new arguments. The original construction packs short arcs on the circle using the Pippenger–Spencer theorem [8] on matchings in hypergraphs, in the qualitative form of Alon and Yuster [2]; Section 3 proves a matching lemma whose constants are explicit in the size of the edges. The original construction also builds an auxiliary polynomial by an iteration whose length was chosen qualitatively; Lemma 4.2 shows that its error decays exponentially, so logarithmically many steps suffice. The remaining steps re-derive the original estimates with numerical constants.

The bound is far too large to be practical, and no polynomial at these lengths has been computed. The three nested exponentials come from the packing step, where the size of the required prime grows exponentially in the number of arcs.

1.1 Prior work and novelty

For complex coefficients of modulus one, Kahane [7] constructed ultraflat polynomials with an explicit rate, and Bombieri and Bourgain [5] improved it. For real signs, Balister, Bollobás, Morris, Sahasrabudhe and Tiba [4] proved that the modulus can be kept between two constant multiples of $\sqrt{N}$, and [9] proved ultraflatness with no rate. Erdélyi [6] states a lower bound for the maximum modulus that is compatible with the rate here. The construction in this paper is that of [9, 10]; the contribution is the explicit bound and the two arguments above.

Quantitative versions of the matching theorem exist. Alon, Kim and Spencer [1] and Kang, Kühn, Methuku and Osthus [3] bound the number of uncovered vertices with explicit exponents. As far as we found, their constants are not given explicitly as functions of the edge size, which is what is needed here because the edge size grows with the accuracy. The lemma in Section 3 is a simple nibble argument written to give such constants; we do not claim it improves on those results.

On October 10, 2026 we searched arXiv (queries on "ultraflat" and on "Littlewood polynomials", through submissions of October 8), the openai/math preprint repository, and the web. We found no earlier explicit rate for real Littlewood polynomials. This is a bounded search, not a proof that none exists.

2 Statement and dependencies

Write $e(t) = \exp(2\pi i t)$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, with normalized Haar measure. All logarithms are natural unless a base is displayed. Set

$$\mathcal{E}_3(x) = \exp(\exp(\exp x)).$$

A real Littlewood polynomial of length N is $P(z) = \sum_{j=0}^{N-1} \epsilon_j z^j$ with $\epsilon_j \in \{-1, 1\}$. Its normalized modulus error is

$$\mathcal{D}(P) = \sup_{|z|=1} \left| \frac{|P(z)|}{\sqrt{N}} - 1 \right|.$$

Theorem 2.1. *For every $0 < \varepsilon < 1$ and every integer*

$$N \geq \mathcal{E}_3\left(10^{12} \log^2 \frac{2}{\varepsilon}\right),$$

there is a real Littlewood polynomial P_N of length N with $\mathcal{D}(P_N) \leq \varepsilon$.

Corollary 2.2. *For every integer $N \geq \mathcal{E}_3(10^{12} \log^2 4)$, there is a real Littlewood polynomial of length N such that*

$$\mathcal{D}(P_N) \leq 2 \exp\left(-10^{-6} \sqrt{\log\log\log N}\right).$$

The signs may be chosen separately for each length.

Proof. Apply Theorem 2.1 with $\varepsilon = 2\exp(-10^{-6}\sqrt{\log\log\log N})$. The stated lower bound on N makes $0 < \varepsilon \leq 1/2$, and substitution in the theorem's threshold gives exactly N. □

The starting architecture is the auxiliary-torus, signed-packing and phase construction of [9, 10]. We record its required quantitative estimates below. The standard external discrepancy input is the partial coloring theorem of Lovett–Meka [11, Theorem 4]; Bertrand's postulate supplies a prime in a prescribed factor-two interval. The matching argument below is self-contained and replaces the qualitative Pippenger–Spencer input in the source construction. No assertion about the running time of a coefficient-generation algorithm is made.

3 An explicit matching lemma

Theorem 3.1. *Let $r \geq 2$, $0 < p \leq 1/2$, $0 < \eta \leq 1/4$, and let*

$$q \geq \left(\frac{2^{20} r}{p\eta}\right)^{328r}.$$

Suppose a finite simple r-uniform hypergraph has $n \leq rq$ vertices and a real reference degree $D \geq p^{r-1}q$. Put $a_0 = (1 + p^{-(r-1)})q^{-1/8}$. If every vertex has degree in $[(1-a_0)D, (1+a_0)D]$ and every pair-codegree is at most $D/(p^{r-1}q)$, then a matching covers at least $(1-\eta)n$ vertices.

We first prove a concentration estimate with its numerical constant.

Lemma 3.2. *Suppose X is a function of m independent Bernoulli(ξ) variables, and changing one variable changes X by at most A. With $V = m\xi(1-\xi)A^2$, for $u > 0$,*

$$\mathbb{P}(|X - \mathbb{E}X| \geq u) \leq 2\exp\left(-\frac{u^2}{2(V + Au/3)}\right).$$

The zero-variance case is interpreted deterministically.

Proof. Expose the variables successively. Each Doob difference is $(Z-\xi)\Delta$, where coupling the same future variables gives $|\Delta| \le A$. Its conditional variance is at most $\xi(1-\xi)A^2$. If Y is a centered difference with $|Y| \le A$, then $j! \ge 2 \cdot 3^{j-2}$ for $j \ge 2$ gives

$$\mathbb{E}(e^{\lambda Y} \mid \text{past}) \le \exp\left(\frac{\lambda^2 \mathbb{E}(Y^2 \mid \text{past})}{2(1-\lambda A/3)}\right), \qquad 0 < \lambda < 3/A.$$

Iterate this inequality and take $\lambda = u/(V + Au/3)$. Apply the same argument to $-X$ for the other tail. □

Proof of Theorem 3.1. Put $k = r-1$ and define

$$L = \log(8/\eta), \qquad \theta = \frac{\eta}{2048 r L}, \qquad T_0 = \lceil 4L/\theta \rceil,$$
$$S = (\eta/8)^{10r}, \qquad E_0 = (8/\eta)^{40r}, \qquad s = \frac{\theta}{256 L E_0},$$
$$B_0 = 2^{32} r^3 L^3 \eta^{-1} p^{-2k} (8/\eta)^{100r}.$$

It suffices initially to assume $q \ge Q_*$, where

$$Q_* = \max\left\{(32p^{-k}E_0)^8, \quad 640 r^2 L E_0 p^{-k} S^{-1}, \quad p^{-k}S^{-1}, \quad (32 r T_0 B_0)^2\right\}. \tag{1}$$

We verify the simpler stated threshold at the end.

Let c denote the maximum pair-codegree in the original hypergraph. The case $n = 0$ is immediate. Otherwise the hypotheses below ensure positive degrees, so $c > 0$. Define deterministic reference degrees by

$$D_0 = D, \qquad \xi_t = \theta/D_t, \qquad \rho_t = (1-\xi_t)^{(r-1)D_t}, \qquad D_{t+1} = D_t \rho_t.$$

At round t, mark each current edge independently with probability ξ_t. Accept each marked edge that meets no other marked edge, and delete every vertex in any marked edge. Continue on the induced surviving hypergraph. Accepted edges from all rounds form a matching.

The inequalities $\theta T_0 \le 5L$ and $\log(1-x) \ge -2x$ for $0 \le x \le 1/2$ imply, inductively,

$$D_t \ge D\exp(-2r\theta t) \ge DS \ge 1, \qquad \rho_t \ge e^{-2r\theta} \ge 1/2. \tag{2}$$

There is no circularity: the bound at time t gives $\xi_t \le \theta < 1/2$ and hence the bound at time $t+1$. Set

$$\kappa = (p^k q S)^{-1}.$$

Pair-codegrees do not increase on taking induced subhypergraphs, and (1) gives

$$\frac{c}{D_t} \le \kappa \le \frac{1}{640 r^2 L E_0}, \qquad a_0 \le \frac{1}{16 E_0}. \tag{3}$$

Conditional degree expectation. Suppose all current degrees are in $(1 \pm a_t)D_t$, with $a_t \le 1/4$. Condition on a vertex v surviving this round. This forces exactly the edges incident to v to have mark zero; every other mark remains an independent Bernoulli(ξ_t) variable. For $e \ni v$, let m_e count edges not containing v that meet $e \setminus \{v\}$. Removing the v-incident edges and subtracting pairwise overlaps gives

$$(r-1)(1-a_t)D_t - \frac{r(r-1)}{2}c \le m_e \le (r-1)(1+a_t)D_t.$$

The conditional survival probability of e is $(1-\xi_t)^{m_e}$. Consequently the conditional mean μ_v of the new degree satisfies

$$(1-a_t)e^{-z_t} \le \frac{\mu_v}{D_{t+1}} \le (1+a_t)e^{z_t}, \qquad z_t = 2\theta(ra_t + r^2c/D_t) \le 1.$$

Using $e^z \le 1+2z$ on $[0,1]$ gives

$$\left|\frac{\mu_v}{D_{t+1}} - 1\right| \le (1+8r\theta)a_t + 8r^2\theta\kappa. \tag{4}$$

Conditional concentration. Changing an unconditioned mark on an edge f changes the new degree of v by at most $\sum_{u\in f}\operatorname{codeg}(v,u) \le rc$. There are at most $2nD_t/r$ marks. Lemma 3.2 applies with $A = rc$ and $V \le 2n\theta rc^2$. Since $D_{t+1} \ge D_t/2$,

$$\begin{aligned}&\mathbb{P}\big(|d_{t+1}(v) - \mu_v| > sD_{t+1} \mid v \text{ survives, current hypergraph}\big)\\ &\quad\le 2\exp\left[-\min\left\{\frac{s^2D_t^2}{32n\theta rc^2}, \frac{sD_t}{8rc}\right\}\right] \le 2e^{-q/B_0}.\end{aligned} \tag{5}$$

Indeed $n \le rq$ and $D_t/c \ge p^k qS$, and

$$\frac{32\theta r^2}{s^2p^{2k}S^2} = B_0, \qquad \frac{8r}{sp^kS} \le B_0.$$

Multiplying the conditional probability by the probability of survival also bounds the probability of a surviving vertex with this deviation.

Closing the induction. Define

$$a_{t+1} = (1+8r\theta)a_t + 8r^2\theta\kappa + s.$$

For every $t \le T_0$, this deterministic recurrence gives

$$a_t \le E_0\left(a_0 + 40r^2L\kappa + \frac{5Ls}{\theta}\right) \le \frac{1}{16} + \frac{1}{16} + \frac{5}{256} < \frac{1}{4}.$$

Thus the hypotheses needed at each good round persist. A union bound over at most nT_0 vertex-round positions shows that the probability of any degree failure is at most $2rqT_0e^{-q/B_0} \le 1/8$. For the last inequality, (1) gives $q/B_0 \ge 32rT_0\sqrt{q}$, whereas

$$\log(16rqT_0) \le (16rT_0+1)\sqrt{q} \le 32rT_0\sqrt{q}.$$

Here $\log q \le \sqrt{q}$ for $q \ge 1$.

Survivors and collision losses. Stop the process on the first degree failure. On a good current history, each vertex survives the next round with probability at most $e^{-3\theta/4}$. Let G_t be the event that all degree conditions hold through time t, and let V_t be the surviving vertex set. Retaining the good-history indicators in conditional expectation gives

$$\mathbb{E}(|V_{t+1}|\mathbf{1}_{G_{t+1}}) \le e^{-3\theta/4}\mathbb{E}(|V_t|\mathbf{1}_{G_t}).$$

It follows that

$$\mathbb{E}(|V_{T_0}|\mathbf{1}_{G_{T_0}}) \le n(\eta/8)^3, \qquad \mathbb{P}(G_{T_0},\ |V_{T_0}| > \eta n/2) \le \eta^2/256 \le 1/4096.$$

Every removed vertex that is not covered by an accepted edge lies in a marked edge having another marked neighbor. The number of such vertices is at most r times the number of ordered intersecting pairs of distinct marked edges. There are at most $2nD_t/r$ edges and at most $2rD_t$ neighbors per edge, so expected waste in one executed round is at most $4nr\theta^2$. Expected total waste is therefore at most $20nr\theta L = 20\eta n/2048$. The probability that it exceeds $\eta n/2$ is less than $1/32$. Together the three failure bounds sum to less than one. Some outcome has at most ηn uncovered vertices.

The stated threshold. Set $W = 2^{20}r/(p\eta) \ge 2^{24}$. Each of r, p^{-1}, η^{-1}, L is at most W, and $8/\eta \le W$, $2^{32} \le W^2$. Hence

$$T_0 \le W^5, \quad E_0 \le W^{40r}, \quad S^{-1} \le W^{10r}, \quad B_0 \le W^{102r+7}.$$

The four terms in (1) are at most W^{328r}, W^{51r+3}, W^{11r-1} and $W^{204r+28}$, respectively. For $r \ge 2$ all are at most W^{328r}, completing the proof. □

4 Quantitative auxiliary data

4.1 The discrepancy input

We use the following explicit form of [9, Lemma 3.2].

Lemma 4.1. *Let $Z_0, \dots, Z_{n-1} \in \mathbb{C}$ satisfy $|Z_j| \le 1$. Write $Z_j = d_j|Z_j|$ with $|d_j| = 1$, taking $d_j = 1$ when $Z_j = 0$, and put $\mu = \frac{1}{2}\sum_j(1 - |Z_j|)$. There are $\eta_j \in \{d_j, -d_j\}$ such that*

$$\sup_{t\in\mathbb{T}} \left| \sum_{j=0}^{n-1} (\eta_j - Z_j)e(jt) \right| \le C_R \left(1 + \sqrt{\mu \log(80n/\mu)}\right), \qquad C_R = 1024. \tag{6}$$

If $\mu = 0$ the error can be zero. For real inputs the outputs are real signs. The same assertion holds on any consecutive interval of frequencies.

Derivation of the numerical constant. The Lovett–Meka partial coloring theorem [11, Theorem 4] allows arbitrary vectors and arbitrary initial points in the cube. Taking the tolerance to zero by compactness gives an exact partial coloring that fixes at least half the active coordinates. For a real $R \times s$ matrix with entries in $[-1, 1]$ and $R \ge s$, use the threshold $4\sqrt{\log(16R/u)}$ when u coordinates remain active. The total discrepancy in a complete coloring is bounded by

$$4\sqrt{sL_0} \sum_{h\ge 0} 2^{-h/2}\sqrt{1 + h/4}, \qquad L_0 = \log(16R/s).$$

The series is less than 7, using $2^{-1/2} < 3/4$ and $\sqrt{1 + h/4} \le 1 + h/4$. Since $L_0 \le 4\log(2R/s)$, a valid complete matrix discrepancy constant is 64.

For completeness, set $p_j = (1 - |Z_j|)/2$ and sample real and imaginary parts of $d_j e(jt)$ on the $20n$ equally spaced points of $\mathbb{T}$. This gives $R = 40n$ real rows. First round p_j down to a dyadic mesh of size 2^{-J} with $n2^{-J} \le 1$, costing at most 1 in row norm. At stage h, color the coordinates with odd numerator on that mesh, reverse all signs if necessary to keep their mass nonincreasing, and

move them by 2^{-h}. Their number s_h is at most $\min(n, 2^h\mu)$. Monotonicity of $s\log(80n/s)$ and the geometric sum $\sum_{h\ge1} 2^{-h/2} = 1+\sqrt{2} < 5/2$ give row error at most $1 + 160\sqrt{\mu\log(80n/\mu)}$. Passing to complex values costs $2\sqrt{2}$. The polynomial derivative bound used in [9, Lemma 3.2] costs at most $(1-\pi\exp(1)/20)^{-1} < 2$ in passing from the grid to the circle. These factors are bounded by the constant in (6). For real inputs all d_j are real signs. A unimodular monomial handles shifted frequency intervals. □

4.2 Exponential saturation of the auxiliary recursion

Let

$$p_0 = 0, \qquad p_j(y_1,\dots,y_j) = p_{j-1} + \frac{1}{2}(1-p_{j-1}^2)\cos(2\pi y_j).$$

These are real trigonometric polynomials, $|p_j| \le 1$, and their means are zero.

Lemma 4.2. *For every integer $d \ge 0$,*

$$1 - \int_{\mathbb{T}^d} p_d^2 \le (7/8)^{d/2}.$$

Proof. Let $H_0(x) = \sqrt{1-x^2}$. Fix $|x| < 1$, write $u = 1-x^2$, and pair $c = \cos(2\pi y)$ with $-c$. Put $A = 1 - uc^2/4$ and $D_0 = 1 - c^2/2 + uc^2/4$. The square of the paired mean of $H_0(x+uc/2)/\sqrt{u}$ is

$$\frac{A + \sqrt{A^2 - x^2c^2}}{2}.$$

Here $D_0 \ge 1/2$ and $D_0^2 - (A^2 - x^2c^2) = (1-u)c^4/4 \ge 0$. The displayed expression is therefore at most $(A+D_0)/2 = 1 - c^2/4$. Jensen's inequality and $\mathbb{E}c^2 = 1/2$ give

$$\mathbb{E}_y H_0\left(x + \tfrac{1}{2}(1-x^2)\cos(2\pi y)\right) \le \sqrt{7/8}\, H_0(x).$$

The endpoints $x = \pm1$ are immediate. Iterate from $p_0 = 0$ and use $H_0^2 \le H_0$. □

4.3 A closed auxiliary envelope

Fix $0 < \delta < 1/100$. The following symbols concern only this section; in particular T is the spreading parameter, not the number of matching rounds. Define

$$\begin{aligned}
&d = \max\left(2, \left\lceil \frac{2\log(1/\delta)}{\log(8/7)} \right\rceil\right), && H_0 = 2^{d-1}, && k_0 = 2^{d+1} - 2,\\
&K = 3^d 2^{d(d-1)/2}, && B_1 = 2H_0 + 2, && G_0 = dH_0 B_1^{d-1},\\
&P_0 = d^2 H_0^2 K, && \tau = \frac{\delta}{4KH_0}, && \alpha = \frac{\delta}{4K\sqrt{K}},\\
&x_0 = \frac{\alpha}{64C_R}, && q_0 = x_0^4, && h_0 = q_0/12,\\
&K_1 = 64, && K_3 = 2^{26}, && M_2 = 1 + 2K_3 h_0^{-3},\\
&E_1 = (2K_1+6)h_0^{-2} + 3h_0^{-3}, && \beta_* && = \frac{\tau 2^{-2k_0}}{G_0 P_0^{K-1}}.
\end{aligned} \tag{7}$$

Finally put

$$R_0 = \max\left\{h_0^{-1}, \frac{1+2\pi M_2}{q_0}, \frac{E_1}{q_0}, 1 + (4C_R/\alpha)^2, \frac{4K}{\delta}\right\}, \qquad T = \lceil R_0/\beta_* \rceil. \tag{8}$$

Proposition 4.3. *There is a real trigonometric polynomial $F(y)=\sum_{a\in A} c_a e(a\cdot y)$ on $\mathbb{T}^m$ and a vector $v\in\mathbb{R}^m$ such that A consists of at least two nonparallel sign pairs, $\lambda_a=a\cdot v\neq 0$, $w_a=|\lambda_a|$, and*

$$\begin{aligned} \|F\|_\infty &\le 1+\delta,\\ \frac{(1-\delta)^3}{(1+\delta)^3} &\le \sum_{a\in A} w_a \le \frac{1}{1+\delta},\\ 1+\delta \le \frac{|c_a|^2}{w_a} &\le \frac{(1+\delta)^3}{(1-\delta)^2}, \qquad 1\le \frac{|c_a|}{\sqrt{w_a}}\le 1+4\delta<2. \end{aligned} \tag{9}$$

The parameters obey

$$\begin{aligned} m &\le d+K/2, & |A| &\le K\big(T(P_0+1)+1\big)^K,\\ \max_{a\in A}\|a\|_\infty &\le \max\{H_0, T(P_0+1)/2\}, & \min_{a\in A} w_a &\ge \frac{1}{\tau T^K(1+\delta)^3}. \end{aligned} \tag{10}$$

All these data are fixed before the final length N is chosen.

Proof. We quantify the construction in [9, Section 4]. Write $p_d=\sum_{s\in\mathcal{S}} b_s e(s\cdot y)$. Every coefficient is an integer multiple of 2^{-k_0}, since the denominator exponents satisfy $k_j=2k_{j-1}+2$. Coordinate degrees satisfy $|s_i|\le 2^{d-i}\le H_0$, and consequently $|\mathcal{S}|\le K$. Every frequency's last nonzero coordinate is ±1, by induction in the displayed recursion. Thus distinct sign pairs are nonparallel. Lemma 4.2 and Parseval give

$$1-\delta\le\sum_s |b_s|^2\le 1, \qquad \sum_s |b_s|\le\sqrt{K}, \qquad |b_s|\ge 2^{-k_0}.$$

Set $\gamma=(1,B_1,\dots,B_1^{d-1})$ and $\rho_s=|s\cdot\gamma|$. Leading-digit domination gives $1\le\rho_s\le G_0$. Choose representatives $s_1,\dots,s_D$, where $2\le D\le K/2$. For each j choose an integer vector $W_j^0\in s_j^\perp$ that is not orthogonal to another representative. Here is a bounded choice. If the last nonzero coordinate of s_j is at i_j, use the integer basis $e_i-(s_j)_i(s_j)_{i_j}^{-1}e_{i_j}$, $i\neq i_j$, and coefficients in $\{0,\dots,D-1\}$. Each of the $D-1$ forbidden hyperplanes contains at most D^{d-2} grid points, fewer in total than D^{d-1}. This gives

$$\big\|W_j^0\big\|_\infty\le dH_0K, \qquad 1\le|s_\ell\cdot W_j^0|\le P_0 \quad (\ell\neq j).$$

Put

$$Q_j=\prod_{\ell\neq j}|s_j\cdot W_\ell^0|, \quad u_j=\frac{|b_{s_j}|^2}{\rho_{s_j}Q_j}, \quad W_j=W_j^0+\tau\frac{u_j}{(s_j)_{i_j}}e_{i_j}.$$

For $\beta_{s,j}=s\cdot W_j$, each off-diagonal factor has relative perturbation at most $\tau H_0\le\delta/(4K)$, and the diagonal factor is exactly τu_j in absolute value. Hence

$$(1-\delta)\tau\frac{|b_s|^2}{\rho_s}\le\prod_{j=1}^D|\beta_{s,j}|\le(1+\delta)\tau\frac{|b_s|^2}{\rho_s}, \qquad \beta_*\le|\beta_{s,j}|\le P_0+1. \tag{11}$$

We next record why (8) suffices for every spreading factor. Take $\kappa_0(x)=C\exp(-1/(1-x^2))$ for $|x|<1$, zero otherwise, normalized to have integral one. Its constant satisfies $C<4$, and direct differentiation gives $\|\kappa_0'\|_1\le K_1$ and $\|\kappa_0'''\|_1\le K_3$. For example the third derivative is

bounded by $4 \cdot 140(1-x^2)^{-6}\exp(-1/(1-x^2))$; the bound $2 \cdot 4 \cdot 140 \cdot 6^6 < 2^{26}$ suffices. Let $g = \mathbf{1}_{[-1/2+2h_0,\,1/2-2h_0]} * \kappa_{0,h_0}$. Then $0 \le g \le 1$, its support stays h_0 from the endpoints,

$$\int g^2 \ge 1 - q_0/2, \quad \|g'\|_1 \le 2, \quad \|g''\|_1 \le 2K_1/h_0, \quad \left\|g^{(4)}\right\|_1 \le 2K_3/h_0^3, \quad \int \xi^2 |\widehat{g}(\xi)|\, d\xi \le M_2.$$

For a nonzero β in the finite list, set $a = T|\beta|$, $L_\beta = 2\lfloor a/2 \rfloor + 1$, and $I_\beta(u) = \int g(y) e(T\beta y^2/2 - uy)\, dy$. The exact quadratic Fourier identity gives

$$\sqrt{L_\beta}|I_\beta(u)| \le \sqrt{1+1/a}(1+\pi M_2/a) \le 1 + (1+2\pi M_2)/a \le 1 + q_0.$$

For $|u| > \lfloor a/2 \rfloor$, put $v_u = |u| - a/2 > 0$. Twice integrating by parts yields

$$|I_\beta(u)| \le \frac{1}{4\pi^2}\left(\frac{\|g''\|_1}{(v_u + ah_0)^2} + \frac{3a\,\|g'\|_1}{(v_u+ah_0)^3} + \frac{3a^2\,\|g\|_1}{(v_u+ah_0)^4}\right).$$

Since $ah_0 \ge 1$, the two-sided sum of $(v_u + ah_0)^{-j}$ is at most $4(ah_0)^{1-j}$ for $j \ge 2$. The exterior coefficient sum is therefore at most $E_1/a \le q_0$. Parseval gives retained energy at least $1 - 2q_0$. Apply Lemma 4.1 to $Z_u = \sqrt{L_\beta} I_\beta(u)/(1+q_0)$. Its normalized defect is at most $2q_0$, so the resulting equal-magnitude polynomial f_β obeys

$$\left\|f_\beta - g(\cdot)e(T\beta(\cdot)^2/2)\right\|_\infty \le C_R L_\beta^{-1/2} + C_R\sqrt{2q_0\log(40/q_0)} + 2q_0 < \alpha.$$

Indeed the three terms are bounded respectively by $\alpha/4$, $\alpha/128$, and $\alpha/4$ from (7)–(8). For the middle term one may use $x_0 \le 1/64$, $\log 80 < 5$, $\log(1/x_0) \le 1/x_0$, and $x_0\sqrt{2(\log 80 + 4\log(1/x_0))} < 1/2$. Choose $f_{-\beta} = \overline{f_\beta}$.

Define, on $\mathbb{T}^{d+D}$,

$$F(y,z) = \sum_{s\in\mathcal{S}} b_s e(s\cdot y) \prod_{j=1}^{D} f_{\beta_{s,j}}(z_j).$$

Replacing every factor by its target gives

$$\left(\prod_j g(z_j)\right) p_d\left(y + \frac{T}{2}\sum_j W_j z_j^2\right),$$

of norm at most one. Telescoping the products costs at most $\sqrt{K}D\alpha(1+\alpha)^{D-1} < \delta$. Each box over s has $n_s = \prod_j L_{\beta_{s,j}}$ frequencies, all with coefficient modulus $|b_s|/\sqrt{n_s}$. Moreover

$$(1-\delta)T^D \prod_j |\beta_{s,j}| \le n_s \le (1+\delta)T^D \prod_j |\beta_{s,j}|,$$

since every factor $L_\beta/(T|\beta|)$ differs from one by at most $\delta/(4K)$. Set

$$v = \left(\frac{\gamma}{\tau T^D(1+\delta)^3}, 0\right).$$

The weight on the box over s is $\rho_s/(\tau T^D(1+\delta)^3)$. Combining the bounds for n_s with (11) proves (9) and (10). Nonparallelism follows by projecting a putative parallel pair to its original s coordinates. □

Remark 4.4. There is no necessary unbounded search for coefficient phase precision. Let $L_{\max} = T(P_0+1)+1$ and $R_{\mathrm{ph}} = \lceil 64\sqrt{L_{\max}}/\alpha \rceil$. Rounding each positive-β unit coefficient to a nearest R_{ph}-th root of unity changes the normalized factor by at most $\alpha/16$. The preceding strict estimate has this much room. Conjugating the negative-β factors preserves all conclusions. The later analytic bounds depend only on moduli and phase derivatives, not on denominators of these arguments.

5 Quantitative signed packing

Let $a_1, \dots, a_r \in \mathbb{Z}^m \setminus \{0\}$ be pairwise nonparallel, $r \geq 2$, and let $w_i > 0$ satisfy $2\sum_i w_i < 1$. We apply the following construction to representatives of the sign pairs in Proposition 4.3. Write

$$\Delta = 1 - 2\sum_i w_i, \qquad \alpha_i = 2w_i + \Delta/r, \qquad M_0 = \lceil 4r(r+1)/\Delta \rceil + 1.$$

Choose L_i to be a nearest positive odd integer to $M_0\alpha_i$, and set

$$L = \sum_i L_i, \qquad p = \frac{1}{2L}, \qquad k = r - 1, \qquad \eta = \frac{\Delta}{16r}.$$

Since $|L_i - M_0\alpha_i| \leq 1$ and $M_0\Delta/r \geq 4(r+1)$,

$$L_i - 2w_iL \geq \frac{3M_0\Delta}{4r}, \qquad 1 - \max_i \frac{2Lw_i}{L_i} \geq \frac{\Delta}{2r} = 8\eta. \tag{12}$$

For the auxiliary data, $\Delta > \delta/2$ and thus $L \leq 64r^2/\delta$, $\eta \geq \delta/(32r)$, and $\eta \leq 1/32$.

Let $b \geq 2$ be the rational rank of the row matrix. Let $A_0 \geq 1$ bound the absolute values of one nonzero b-minor and one nonzero 2-minor for every pair of rows. Put

$$C_{mmom} = (24kL + 18k^2)p^{-2k}.$$

Choose a prime q strictly greater than all the following quantities:

$$4, \quad A_0, \quad [2r(C_{\mathrm{mom}} + 3k(k-1) + 1)]^8, \quad [(1+8L)/\eta]^8, \quad [2^{20}r/(p\eta)]^{2^{20}r^2}. \tag{13}$$

The last term is a convenient larger threshold than the one in Theorem 3.1.

Proposition 5.1. *For this q, there are an integer $1 \leq H \leq q$ and points $\theta_1, \dots, \theta_H \in q^{-1}\mathbb{Z}^m/\mathbb{Z}^m$ such that the signed closed arcs centered at $\pm a_i \cdot \theta_h$, of lengths w_i/H, are pairwise disjoint. The arcs have pairwise separation at least $6\eta/q$, and their distances from 0 and $1/2$ are at least η/q. Moreover*

$$H \geq (1-\eta)^2\frac{q}{2L} \geq (1-2\eta)\frac{q}{2L}.$$

Proof. We quantify the finite-field construction in [10, Section 3]. The choice $q > A_0$ preserves the full rank b and rank two for every row pair over $\mathbb{F}_q$. Put $R = \lfloor\sqrt{q}\rfloor$, $n_+ = (q-1)/2$, and divide the positive sites $\{1, \dots, n_+\}$ into chunks of R consecutive sites, with a possibly shorter final chunk. Their number K_c is at most $3\sqrt{q}/2$. In each chunk independently shift a periodic pattern of consecutive slots of lengths $L_1, \dots, L_r$ by a uniform residue modulo L and retain only full slots. Each retained slot is independently oriented with a fair sign. Since L_i is odd its center is an integer site. The number of type-i slots in a chunk of u sites differs from u/L by at most 2. Thus every realization has between $n_+/L - 2K_c$ and $n_+/L + 2K_c$ slots of each type.

For $s \in \mathbb{F}_q$, let $|s|_q$ be the absolute value of its representative in $[-n_+, n_+]$, and let $C_i(s)$ indicate that s is an oriented center of type i. A site is interior if it is at distance at least L from both ends of its chunk. For interior folded sites, $\mathbb{E}C_i(s) = p = 1/(2L)$. At most $2LK_c$ positive sites are noninterior. An edge is a compatible tuple of one slot of each type, meaning that $a_i \cdot y = s_i$ for all i has a solution in $\mathbb{F}_q^m$. Each tuple is included once, making a simple r-uniform hypergraph.

Fix a candidate center $s \neq 0$ of type i and condition on its presence. This affects only the chunk of $|s|_q$. Exclude centers s for which some $j < \ell$, both different from i, satisfy $a_j \pm a_\ell = ca_i$ with $c \neq 0$ and $|cs|_q \leq R$. There are at most

$$2\binom{k}{2}(2R+1) \leq 3k(k-1)\sqrt{q}$$

such centers for each type. Pairwise rank two excludes a zero constant combination. For a nonexceptional s, sample y uniformly from $a_i \cdot y = s$. Each other evaluation is uniform; the probability that one evaluation is zero, noninterior or in the conditioned chunk is at most

$$\frac{1 + 4LK_c + 2R}{q} \leq \frac{6L+3}{\sqrt{q}}.$$

Two evaluations in the same folded chunk have sum or difference of folded size at most R. A nonconstant combination is uniform, and each sign costs at most $(2R+1)/q \leq 3/\sqrt{q}$. Constant within-copy combinations were excluded above. Across two independent fiber samples, the relevant evaluations are independent and uniform, including when the row indices agree. For $u = 1, 2$ copies, a union bound therefore gives

$$F_u(q) \leq \frac{uk(6L+3uk)}{\sqrt{q}} \tag{14}$$

for failure to occupy distinct interior unconditioned chunks.

Define

$$Z_{i,s} = q^{-(m-1)} \sum_{a_i \cdot y = s} \prod_{j \neq i} C_j(a_j \cdot y).$$

On good sample tuples the center events involve distinct chunks, so their conditional probability is exactly p^{uk}. It follows that

$$|\mathbb{E}Z_{i,s} - p^k| \leq F_1, \quad |\mathbb{E}Z_{i,s}^2 - p^{2k}| \leq F_2, \quad \mathbb{E}(Z_{i,s} - p^k)^2 \leq \frac{24kL + 18k^2}{\sqrt{q}}.$$

All expectations are conditional on the candidate vertex being present. Chebyshev with relative error $q^{-1/8}$ bounds its failure probability by $C_{\mathrm{mom}} q^{-1/4}$. There are at most $r(q-1)$ candidates. Together with the exceptional centers, the expected number of deletions is at most

$$rC_{\mathrm{mom}} q^{3/4} + 3rk(k-1)\sqrt{q} < q^{7/8},$$

by (13). Fix a realization with fewer than $q^{7/8}$ deletions.

The evaluation map has q^{m-b} preimages per compatible tuple. A present vertex therefore has degree $q^{b-1} Z_{i,s}$, and fixing two different types gives pair-codegree at most q^{b-2}. Same-type pair-codegrees are zero. Deletion loses fewer than $q^{7/8} q^{b-2}$ incident edges at each remaining vertex. With $D = p^k q^{b-1}$, the remaining degrees consequently lie in

$$\left(1 \pm (1 + p^{-k}) q^{-1/8}\right) D,$$

and the maximum pair-codegree is at most $D/(p^k q)$. There are at most rq vertices and $D \geq p^k q$. All hypotheses of Theorem 3.1 hold.

The remaining vertex count v obeys

$$\frac{v}{r} \geq \frac{q}{2L} - \frac{1}{2L} - 2K_c - q^{7/8} > (1-\eta)\frac{q}{2L}.$$

Indeed its relative deficit is at most $(1+8L)q^{-1/8} < \eta$. The matching supplied by Theorem 3.1 has the asserted size H, and $H \le q$. Choose one generating vector y_h for each edge and put $\theta_h = y_h/q$. By (12),

$$\frac{L_i}{q} - \frac{w_i}{H} \ge \frac{6\eta}{1-2\eta}\frac{L_i}{q} \ge \frac{6\eta}{q}.$$

Thus each desired arc is inset by at least $3\eta/q$ from its slot boundary. Slot interiors are disjoint, and the positive slots lie in $[1/(2q), 1/2]$. Reflection gives the signed statement and the claimed boundary distances. □

6 Uniform analytic estimates

We give a numerical stationary-phase certificate before using it in the wave construction. These estimates quantify [9, Sections 2 and 5].

Lemma 6.1. *Let I have length $\ell \le 1$, and let ϕ be real C^7 on a neighborhood of I, with fixed sign of curvature. Suppose*

$$0 < b \le \min(1, \inf_I |\phi''|), \qquad B \ge \max\left(1, \max_{2\le j\le 7} \left\|\phi^{(j)}\right\|_\infty\right).$$

Let $f \in C^4$ have support at distance at least $h > 0$ from the endpoints, and let $F_j \ge \left\|f^{(j)}\right\|_\infty$ for $0 \le j \le 4$. Define

$$a_0 = B, \qquad a_k = \frac{B + \sum_{j=1}^{k-1}\binom{k}{j} a_j a_{k-j}}{2b} \qquad (1 \le k \le 5),$$

$$Y_k = \ell a_k + k a_{k-1} \qquad (1 \le k \le 5),$$

$$R_0 = B/b, \qquad R_k = (B/b)\sum_{j=0}^{k-1}\binom{k}{j} R_j Y_{k-j+1} \qquad (1 \le k \le 4),$$

$$V_{0,k} = \sum_{j=0}^{k}\binom{k}{j} F_j R_{k-j} \qquad (0 \le k \le 4),$$

$$V_{a,k} = \sum_{j=0}^{k}\binom{k}{j} R_j V_{a-1,k-j+1} \qquad (1 \le a \le 4, 0 \le k \le 4-a),$$

$$C_M = 4\ell\big(F_0 + (B/b)V_{4,0}\big), \qquad C_{NS} = \ell\left(\frac{F_1}{bh} + \frac{F_0 B}{b^2 h^2}\right).$$

For every real x and $N \ge 1$, the scaled integral $\sqrt{N}\int_I f(t)e(N(\phi(t)-xt))\,dt$ differs from its usual stationary-phase main term by at most $C_M/N + C_{NS}/\sqrt{N}$. If $\phi'(u) = x$ for $u \in I$, that term is

$$\frac{f(u)}{\sqrt{|\phi''(u)|}} e\big(N(\phi(u)-xu) + \operatorname{sgn}(\phi'')/8\big);$$

otherwise it is zero.

Proof. For $x = \phi'(u)$ put $d = \operatorname{sgn}(\phi'')$ and

$$A(t,u) = 2d\int_0^1 (1-s)\phi''(u + s(t-u))\,ds, \quad a(t,u) = \sqrt{A(t,u)}, \quad y = (t-u)a(t,u).$$

Then $b \le A \le B$, its first five t-derivatives are bounded by B, and the phase becomes $\phi(u) - xu + dy^2/2$ exactly. Differentiating $a^2 = A$ and then $y = (t-u)a$ gives the bounds a_k, Y_k above. Also $b/\sqrt{B} \le y' \le B/\sqrt{b}$, so both y' and $1/y'$ are bounded by B/b. Writing $r = 1/y'$, differentiation of $ry' = 1$ gives the R_k bounds. The transformed amplitude $g(y) = f(t(y))r(t(y))$ satisfies $\|g\|_1 \le \ell F_0$ and $\|g^{(4)}\|_1 \le \ell(B/b)V_{4,0}$, since $d/dy = r\, d/dt$. It extends by zero as a compactly supported C^4 function.

The exact quadratic Fourier identity, obtained for instance with a Gaussian regulator, gives error at most $\pi N^{-1} \int \xi^2 |\widehat{g}(\xi)|\, d\xi$. Four integrations by parts show

$$\int \xi^2 |\widehat{g}(\xi)|\, d\xi \le \tfrac{2}{3}\, \|g\|_1 + \frac{2}{(2\pi)^4} \left\|g^{(4)}\right\|_1 \le \|g\|_1 + \left\|g^{(4)}\right\|_1 .$$

This gives C_M/N. For x outside $\phi'(I)$, the derivative $\phi' - x$ has modulus at least bh on the support. A single integration by parts gives $C_{NS}/\sqrt{N}$. Endpoints of the derivative range are covered by the first argument. □

6.1 Main waves and their coherence

Apply Proposition 5.1 to the sign-pair representatives of A from Proposition 4.3. The centers can be written $t^*_{a,h} = -a \cdot \theta_h \pmod 1$, since both signs are packed. Write $m_a = |c_a|/\sqrt{w_a} \in [1, 1+4\delta]$ and set

$$J_h = \left[\frac{h-1}{H} + \frac{\delta}{16H}, \frac{h}{H} - \frac{\delta}{16H}\right], \quad e_0 = \frac{\delta}{32H}, \quad \sigma = \frac{\delta}{24H|A|}. \tag{15}$$

Use the monotone C^6 cutoff

$$\zeta(s) = 12012 \int_0^s t^6(1-t)^6\, dt \quad (0 \le s \le 1),$$

extended by zero and one. For distance d_x to the nearest endpoint of J_h, set $\chi_h = \sigma$ for $d_x \le 2e_0$, interpolate to one using $\zeta((d_x - 2e_0)/e_0)$ on $[2e_0, 3e_0]$, and keep it one thereafter. Set $G_h = 0$ for $d_x \le e_0/2$, interpolate to one on $[e_0/2, e_0]$ using ζ, and keep it one thereafter. Both are constant near the midpoint, so this definition introduces no singularity there. They satisfy

$$\sigma \le \chi_h \le 1, \qquad \int_{J_h} \chi_h^2 \ge \frac{1 - 5\delta/16}{H} > \frac{1-\delta}{H}.$$

Let $Q_h'' = \chi_h^2$ with $Q_h = Q_h' = 0$ at the midpoint. Define

$$\begin{aligned} T_{a,h}(x) &= t^*_{a,h} - \lambda_a Q_h'(x), & X_{a,h} &= T_{a,h}^{-1}, \\ \psi_{a,h}(t) &= (t - t^*_{a,h}) X_{a,h}(t) + \lambda_a Q_h(X_{a,h}(t)). \end{aligned} \tag{16}$$

The image $I_{a,h} = T_{a,h}(J_h)$ is strictly inside its packed arc. The identities

$$\psi'_{a,h} = X_{a,h}, \qquad \psi''_{a,h} = -\frac{1}{\lambda_a \chi_h(X_{a,h})^2}$$

follow by differentiation. On $I_{a,h}$ define

$$B_N(t) = \frac{c_a}{\sqrt{w_a}} e\big(\operatorname{sgn}(\lambda_a)/8 + N\psi_{a,h}(t)\big). \tag{17}$$

The opposite frequency gives the conjugate reflection. Main intervals have total length at least $(1-\delta)\sum_a w_a$.

Apply Lemma 6.1 with the cutoff $G_h(X_{a,h})$ and $x = k/N$. Its leading scaled contribution is, for $x \in J_h$,

$$c_a G_h(x)\chi_h(x)e\big(-kt^*_{a,h} + N\lambda_a Q_h(x)\big).$$

At most one J_h contains x. Summing over a for that h gives

$$G_h(x)\chi_h(x)F\big(k\theta_h + NvQ_h(x)\big), \tag{18}$$

whose modulus is at most $1+\delta$. The congruence defining t^* is legitimate here because k is an integer.

The remaining two endpoint pieces lie where $\chi_h = \sigma$ and have constant curvature of modulus $1/(w_a\sigma^2)$. The primitive of $e(\pm y^2/2)$ has differences bounded by 3 on arbitrary intervals; split at ± 1 and integrate the tails by parts to see this. Including amplitude variation, each endpoint contributes at most $6|c_a|\sigma$ to a scaled Fourier coefficient. The sum over both endpoints and all (a,h) is at most $12H\sigma\sum_a |c_a| \le \delta$.

6.2 Joining the gaps with numerical separation

On $[0,1/2]$, let the gaps have lengths ℓ_j, number $M = rH+1$, and total length L_g. Proposition 5.1 and the total main length give

$$\ell_j \ge \eta/q, \qquad M \le rq+1, \qquad 0 < L_g \le 3.5\delta. \tag{19}$$

Prescribe endpoint derivatives x_z from the adjacent main waves, and use $x_0 = x_{1/2} = 1/8$ at the boundary. Assign to gap j an interval $[P_j, R_j]$ of length $\ell_j/(4L_g)$. Arrange these successively starting at $5/16$, separated by spaces $s_g = 1/[8(M+1)]$. They have total length $1/4$, total spacing less than $1/8$, and lie below $11/16$.

There are at most $4M^2$ forbidden common translations that put one of the $2M$ assigned endpoints on one of at most $2M$ prescribed values. Take the $4M^2+1$ translations

$$\frac{j}{256M^2}, \qquad 0 \le j \le 4M^2.$$

Each forbidden value is within $1/(768M^2)$ of at most one candidate. One candidate remains, and after that translation

$$|P_j - x_z|, |R_j - x_z| \ge \frac{1}{768M^2} \quad \text{for every prescribed } x_z. \tag{20}$$

The intervals still lie in $(1/4, 3/4)$. Their closed ρ-neighborhoods are disjoint for $\rho = 1/[32(M+1)]$.

On each gap choose a continuous piecewise linear leading derivative interpolating $x_{\text{left}} \longrightarrow P_j \longrightarrow R_j \longrightarrow x_{\text{right}}$. Give its outer pieces lengths

$$u_j = \min\left\{\frac{\ell_j}{8}, |P_j - x_{\text{left}}|\left(\frac{\delta}{400M}\right)^2\right\} \tag{21}$$

and the analogous expression on the right. Their nonzero curvatures Λ satisfy $\sum_{\text{outer}} |\Lambda|^{-1/2} \le \delta/200$ on the positive half. The middle curvature is positive and at least $1/(4L_g)$. Integrate to obtain a C^1, piecewise quadratic phase ϕ_j.

Interpolate the endpoint moduli linearly by m_j, prescribing modulus one at 0 and $1/2$. Choose an affine $\gamma_{j,N}$ with both endpoint values in $[0,1)$ so that

$$B_N(t) = m_j(t)e\big(N\phi_j(t) + \gamma_{j,N}(t)\big)$$

matches the two required endpoint values. This is always possible by choosing representatives of the two phase congruences. It gives $|\gamma'_{j,N}| \le 1/\ell_j$. Define the negative half by conjugate reflection. Then B_N is continuous on $\mathbb{T}$, with

$$B_N(-t) = \overline{B_N(t)}, \qquad 1 \le |B_N(t)| \le 1 + 4\delta, \qquad B_N(0) = B_N(1/2) = 1. \tag{22}$$

Leading phase derivatives match at every join, including the periodic endpoints. All lie in $[\delta/(16H), 1 - \delta/(16H)]$.

On each gap piece, the amplitude $f_N = m_j e(\gamma_{j,N})$ satisfies $\|f_N\|_\infty + \int |f'_N| \le 3 + 4\pi < 16$. The quadratic estimate gives scaled outer cost below δ, including reflections. At most one assigned middle derivative interval is within ρ of $x = k/N$. Its piece and reflection cost at most $192\sqrt{L_g}$. Every other pair is nonstationary and costs at most $96/(\rho\sqrt{N})$, by one integration by parts with monotone derivative.

Let $C_{M,\Sigma}$ and $C_{NS,\Sigma}$ be the sums of the constants in Lemma 6.1 over the main cutoff integrals. Put

$$N_{\mathrm{coeff}} = \max\left\{1, \left\lceil \frac{2C_{M,\Sigma}}{\delta} \right\rceil, \left\lceil \left(\frac{2C_{NS,\Sigma}}{\delta}\right)^2 \right\rceil, \left\lceil \left(\frac{96M}{\rho\delta}\right)^2 \right\rceil \right\}. \tag{23}$$

For $N \ge N_{\mathrm{coeff}}$, coherence, endpoints and gaps give

$$\sqrt{N}\,|\widehat{B}_N(k)| \le 1 + 5\delta + 192\sqrt{3.5\delta} < 1 + 500\sqrt{\delta}, \qquad 0 \le k < N. \tag{24}$$

6.3 A summable exterior tail

Partition all main intervals, the three pieces of each gap and their reflections. On a piece p, write $B_N = f_{p,N}e(N\phi_p)$, and let $F_{p,j}$ bound $\left\|f^{(j)}_{p,N}\right\|_\infty$ for $j \le 2$ and $B_{p,j}$ bound $\left\|\phi^{(j)}_p\right\|_\infty$ for $j = 2, 3$. Set $c = \delta/(32H)$. For an exterior integer $k < 0$ or $k \ge N$, $|N\phi'_p - k| \ge c(N + |k|)$. Define

$$\begin{aligned} A_p &= F_{p,1}/c^2 + F_{p,0}B_{p,2}/c^3, \\ A'_p &= F_{p,2}/c^2 + 3F_{p,1}B_{p,2}/c^3 + F_{p,0}B_{p,3}/c^3 + 3F_{p,0}B^2_{p,2}/c^4, \\ K_{\mathrm{tail}} &= 2\sum_p (2A_p + \ell_p A'_p). \end{aligned} \tag{25}$$

The first integration-by-parts boundary terms cancel, by continuity of B_N and matching of the leading derivatives. They also cancel at the periodic endpoints, since $B_N = 1$, $\phi' = 1/8$, and k is an integer. The second integration involves, with $V = N\phi' - k$,

$$\frac{f'}{V^2} - \frac{fV'}{V^3}, \qquad \frac{f''}{V^2} - \frac{3f'V'}{V^3} - \frac{fV''}{V^3} + \frac{3f(V')^2}{V^4}.$$

Thus (25) gives

$$\sum_{k<0 \text{ or } k\ge N} |\widehat{B}_N(k)| \le K_{\mathrm{tail}}/N. \tag{26}$$

One uses $\sum_{k<0 \text{ or } k\ge N}(N + |k|)^{-2} \le 2/N$. The Fourier series is absolutely convergent and equals B_N by continuity and uniqueness. Conjugate symmetry makes every coefficient real.

7 An explicit length threshold

We now bound every datum used in Section 6 by a fixed power of one explicit scalar. This also specifies a threshold smaller than the simpler triple exponential in Theorem 2.1.

For $0 < \varepsilon < 1$ set

$$m_\varepsilon = \left\lceil 2\log_2(2^{20}/\varepsilon) \right\rceil, \qquad \delta = 2^{-4m_\varepsilon}. \tag{27}$$

Using (7)–(8), define

$$\begin{aligned}
\overline{R} &= \left\lceil K\big(T(P_0+1)+1\big)^K \right\rceil, & \overline{m} &= \lceil d + K/2 \rceil, \\
\overline{U} &= \lceil \max\{H_0, T(P_0+1)/2\} \rceil, & w_* &= \frac{1}{\tau T^K (1+\delta)^3}, \\
\overline{L} &= \left\lceil 64\overline{R}^2/\delta \right\rceil, & \overline{p} &= \frac{1}{2\overline{L}}, \\
\overline{\eta} &= \frac{\delta}{32\overline{R}}, & \overline{C} &= (24\overline{R}\,\overline{L} + 18\overline{R}^2)\overline{p}^{-2\overline{R}}, \\
\overline{A} &= \overline{m}!\,\overline{U}^{\overline{m}}.
\end{aligned} \tag{28}$$

Put

$$\begin{aligned}
\overline{Q} = \max\Bigg\{ & 4, \overline{A}, [2\overline{R}(\overline{C} + 3\overline{R}^2 + 1)]^8, [(1+8\overline{L})/\overline{\eta}]^8, \\
& \left[\frac{2^{20}\overline{R}}{\overline{p}\,\overline{\eta}}\right]^{2^{20}\overline{R}^2} \Bigg\},
\end{aligned} \tag{29}$$

$$\begin{aligned}
\overline{q} &= 2\left\lceil \overline{Q} \right\rceil, \\
Z &= 2^{64}\big(\overline{q} + \overline{R} + \delta^{-1} + w_*^{-1} + \overline{\eta}^{-1} + 1\big), \\
N_*(\varepsilon) &= \left\lceil Z^{10^7} \right\rceil.
\end{aligned} \tag{30}$$

Every symbol on the right is an explicit function of ε. In particular, no instance-specific minimum separation or unspecified existence threshold occurs in these formulas.

7.1 All packing choices fit the envelope

For the actual auxiliary support, write $|A| = 2r$. Then $r \le \overline{R}$, $m \le \overline{m}$, and all frequency coordinates are at most $\overline{U}$ in absolute value. Every relevant nonzero minor has absolute value at most $\overline{m}!\,\overline{U}^{\overline{m}}$. The bounds in Section 5 give $L \le \overline{L}$, $p \ge \overline{p}$, and $\eta \ge \overline{\eta}$. Therefore every threshold in (13) is at most $\overline{Q}$. Bertrand's postulate gives a prime

$$\left\lceil \overline{Q} \right\rceil < q \le 2\left\lceil \overline{Q} \right\rceil = \overline{q}.$$

Proposition 5.1 applies to this prime. All corresponding centers have denominator q, $H \le q$, and (19) holds. The vector θ_h can be obtained by solving its finite-field linear system; this step introduces no larger denominator.

7.2 Fixed powers control the analytic constants

From the definition of Z we may use the following loose inequalities:

$$\begin{aligned} H, |A|, q, r, \delta^{-1},\ w_*^{-1},\ \eta^{-1} \le Z, \quad M \le Z^2, \quad \rho^{-1} \le Z^3, \\ \sigma^{-1} \le Z^4, \quad e_0^{-1} \le Z^3, \quad (w_*\sigma^2)^{-1} \le Z^9, \quad \ell_j^{-1} \le Z^2. \end{aligned} \tag{31}$$

The sums of the absolute coefficients of ζ and its first six derivatives are less than 2^{50}. This follows directly by expanding $12012 \int_0^s t^6(1-t)^6\,dt$ and differentiating. Thus

$$\left\|\chi_h^{(j)}\right\|_\infty \le Z^{3j+1}, \qquad \left\|G_h^{(j)}\right\|_\infty \le Z^{4j+1} \quad (1 \le j \le 6).$$

Leibniz's rule bounds the first six derivatives of $T_{a,h}$ by Z^{21}.

Let $\mathcal{B}_{n,k}$ be the exponential partial Bell polynomial. Inverse differentiation gives bounds for $X = T_{a,h}^{-1}$ through

$$\begin{aligned} J_1 &= (w_*\sigma^2)^{-1}, \\ J_n &= (w_*\sigma^2)^{-1} \sum_{k=2}^{n} D_k \mathcal{B}_{n,k}(J_1, \dots, J_{n-k+1}), \qquad 2 \le n \le 6, \end{aligned}$$

where $D_k \le Z^{21}$ bounds $T_{a,h}^{(k)}$. These imply

$$J_n \le Z^{64n^2}, \qquad 1 \le n \le 6. \tag{32}$$

Indeed a Bell monomial of weight n and $k \ge 2$ factors has sum of squared derivative orders at most $(n-k+1)^2 + k - 1 \le n^2 - 2$. The saved 128 exponent units absorb at most 30 from the prefactor and D_k, and at most one from the finite coefficient sum. At least 97 units of slack remain. The case $n = 1$ follows from (31).

For a main cutoff integral in Lemma 6.1, we can take

$$b = 1, \qquad B = Z^{2500}, \qquad F_j = Z^{1500}\ (0 \le j \le 4), \qquad h^{-1} \le Z^{20}. \tag{33}$$

Here $w_a < 1$ and $\chi_h \le 1$ give $|\psi''| \ge 1$; the phase derivative bound follows from J_6. For the amplitude $c_a w_a^{-1/2} e(\mathrm{sgn}(\lambda_a)/8) G_h(X)$, the Bell monomials have weight at most four, giving an exponent below $64 \cdot 4^2 + 17 + 2 < 1500$. The support is at distance at least $w_*\sigma^2 e_0/2$ from each main endpoint, giving the stated h bound.

Substitution into the finite recurrences of Lemma 6.1 gives the following conservative bookkeeping bounds:

$$\begin{aligned} &a_k \le Z^{100000} && (0 \le k \le 5), && Y_k \le Z^{100002} && (1 \le k \le 5), \\ &R_k \le Z^{500000} && (0 \le k \le 4), && V_{0,k} \le Z^{501501}, \\ &V_{a,k} \le Z^{501501+500001a} && (a \le 4), && C_M \le Z^{3000000}, \quad C_{NS} \le Z^{5000}. \end{aligned}$$

For example the five a_k steps at worst double the preceding exponent and add a constant, starting at 2500. Each R_k step adds at most $2500 + 100002 + 1$. The V bounds follow by direct multiplication and one extra power for each finite sum. There are at most $|A|H \le Z^2$ main integrals. Formula (23) therefore yields

$$N_{\mathrm{coeff}} \le Z^{4000000}. \tag{34}$$

For gap pieces, (20) bounds each inverse endpoint difference by $768M^2 \le Z^5$. Formula (21) then bounds the inverse outer lengths by Z^{13}. All gap curvatures are at most Z^{13}; for a middle piece use its length at least $3\ell_j/4$. The gap amplitudes have the uniform derivative bounds

$$F_0 = 2, \qquad F_1 \le 17/\ell_j, \qquad F_2 \le 144/\ell_j^2,$$

which are at most Z, Z^3, Z^5. Main amplitudes are constant, and their phase derivatives through order three are at most Z^{2500}. Since $c^{-1} \le Z^3$ and there are at most Z^3 pieces, direct substitution in (25) gives

$$K_{\mathrm{tail}} \le Z^{6000}, \qquad K_{\mathrm{tail}}/\delta \le Z^{6001}. \tag{35}$$

Thus both analytic thresholds are bounded by $N_*(\varepsilon)$.

7.3 Projection and rounding to real signs

For $N \ge N_*(\varepsilon)$ put

$$S_\delta = 1 + 500\sqrt{\delta}, \qquad Y_k = \frac{\sqrt{N}\,\widehat{B}_N(k)}{S_\delta} \quad (0 \le k < N), \qquad U_Y(t) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} Y_k e(kt).$$

By (24) the Y_k are real and lie in $[-1, 1]$. Equations (26) and (35) give $\|U_Y - B_N/S_\delta\|_\infty \le \delta$. Consequently Parseval implies

$$\frac{1}{N}\sum_k Y_k^2 \ge (S_\delta^{-1} - \delta)^2, \qquad \frac{\mu}{N} = \frac{1}{2N}\sum_k (1 - |Y_k|) \le \frac{1 - S_\delta^{-2}}{2} + \delta \le 501\sqrt{\delta} =: q_\delta < \frac{1}{2}.$$

Lemma 4.1, monotonicity of $q\log(80/q)$ on $(0, 1/2]$, and $\log(80/q_\delta) < 2m_\varepsilon$ give real signs with

$$\begin{aligned} \|U_\epsilon - U_Y\|_\infty &\le C_R\left(N^{-1/2} + \sqrt{q_\delta \log(80/q_\delta)}\right) \\ &\le C_R N^{-1/2} + 32 C_R m_\varepsilon 2^{-m_\varepsilon}, \qquad U_\epsilon(t) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \epsilon_k e(kt). \end{aligned}$$

Set $x = 2^{20}/\varepsilon$. Then $m_\varepsilon \le x$ and $2^{-m_\varepsilon} \le x^{-2}$, so

$$32 C_R m_\varepsilon 2^{-m_\varepsilon} \le \varepsilon/32.$$

Each of $500 \cdot 2^{-2m_\varepsilon}$ and $5 \cdot 2^{-4m_\varepsilon}$ is also at most $\varepsilon/32$. Moreover $(2C_R/\varepsilon)^2 \le \delta^{-1} \le Z \le N$. Using (22), the lower modulus deficit is at most $500\sqrt{\delta} + \delta$ plus the rounding error, and the upper excess is at most 5δ plus that error. Both are therefore at most

$$500 \cdot 2^{-2m_\varepsilon} + 5 \cdot 2^{-4m_\varepsilon} + 32 C_R m_\varepsilon 2^{-m_\varepsilon} + C_R/\sqrt{N} \le \varepsilon.$$

This proves the desired two-sided estimate whenever $N \ge N_*(\varepsilon)$. All parameters other than the harmless affine endpoint corrections were fixed before N was chosen. There is no divisibility restriction on N.

8 The triple exponential bound

It remains to bound (30) by the threshold in Theorem 2.1. Put $t = \log(1/\delta)$. The precision choice (27) ensures $t \geq 160 \log 2 > 110$ for the entire range $0 < \varepsilon < 1$. Directly from the auxiliary formulas,

$$\begin{aligned} d &\leq 16t, & \log K &\leq 110t^2, & \log P_0 &\leq 150t^2, \\ \log G_0 &\leq 250t^2, & \log \tau^{-1} &\leq 130t^2, & \log \alpha^{-1} &\leq 170t^2, \\ \log R_0 &\leq 3000t^2, & \log \beta_*^{-1} &\leq e^{130t^2}, & \log T &\leq e^{200t^2}. \end{aligned}$$

Here are details of the larger terms. The definitions give $\log x_0^{-1} \leq 171t^2$, $\log q_0^{-1} \leq 684t^2$, and $\log h_0^{-1} \leq 685t^2$. Thus $\log M_2, \log E_1 \leq 2056t^2$, which bounds the maximum defining R_0 as displayed. In

$$\log \beta_*^{-1} = \log \tau^{-1} + 2k_0 \log 2 + \log G_0 + (K-1) \log P_0,$$

the last term is at most $150t^2 e^{110t^2}$ and absorbs the others into e^{130t^2}. Taking the ceiling in T is absorbed by e^{200t^2}.

Expanding (28) now yields

$$\begin{aligned} \log \max(2, \overline{R}, \overline{m}, \overline{U}, w_*^{-1}, \overline{A}) &\leq e^{500t^2}, \\ \log \max(2, \overline{L}, \overline{p}^{-1}, \overline{\eta}^{-1}) &\leq e^{510t^2}. \end{aligned}$$

For example $\log(T(P_0+1)+1) \leq e^{201t^2}$, so $\log \overline{R} \leq e^{312t^2}$. Also $\overline{m} \leq K$, and $\log(\overline{m}!) \leq \overline{m} \log \overline{m}$ handles the factorial in $\overline{A}$.

The logarithms of $\overline{C}$ and of the matching term in (29) are bounded by $\exp(\exp(600t^2))$. Their potentially largest contributions are respectively

$$2\overline{R} \log \overline{p}^{-1}, \qquad 2^{20} \overline{R}^2 \log \left(\frac{2^{20} \overline{R}}{\overline{p}\,\overline{\eta}} \right).$$

The remaining finite sums, powers and ceilings in (29)–(30) fit, with room, within

$$N_*(\varepsilon) \leq \mathcal{E}_3(1000t^2). \tag{36}$$

These estimates use $t > 110$ throughout, so no unspecified small-error cutoff is needed. Finally,

$$m_\varepsilon \leq 41 + 2\log_2(1/\varepsilon), \qquad t \leq 164 \log 2 + 8 \log(1/\varepsilon) \leq 172 \log(2/\varepsilon).$$

Since $1000 \cdot 172^2 = 29{,}584{,}000 < 10^{12}$, (36) proves Theorem 2.1.

Scope

The matching estimate is proved here, including the concentration inequality it needs. All gap separations and analytic thresholds are given by explicit numerical constants. The external results used are the Lovett–Meka partial-coloring theorem and Bertrand's postulate, both cited rather than re-proved. The triple exponential bounds the length that suffices; it says nothing about the running time of any algorithm. No polynomial at these lengths has been computed. The signs may differ from one length to the next; the paper does not construct a single infinite sequence of signs.

AI assistance

The arguments in this paper were produced with the assistance of AI models and checked by separate AI proof audits. Agreement between AI audits is not independent human verification, and the paper has not been peer reviewed.

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